

**The Impact of Improved Water and Sanitation on Child Nutritional
Outcomes
in Sub-Saharan Africa: Evidence from DHS and Geospatial Data**

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Abstract

(Written after all other chapters, to match the balanced evidence actually presented in Sections 5–7, rather than anticipating it.)

This thesis examines associations between household and community water and sanitation conditions and child anthropometric outcomes (height-for-age, weight-for-age, weight-for-height) using Demographic and Health Survey data from four Sub-Saharan African countries: Ethiopia, Ghana, Kenya, and Nigeria. It distinguishes throughout between a child’s own household facilities and a leave-one-out measure of improved-facility coverage among other households in the same survey cluster, estimating each under the fixed-effects structure its own construction requires. Household water and sanitation status show no statistically significant association with any of the three outcomes once comparisons are restricted to households within the same cluster. Community-level coverage shows a large, significant unadjusted association that attenuates to statistical insignificance once socioeconomic status and region fixed effects are added, a pattern consistent with, though not conclusive evidence of, confounding by local socioeconomic conditions, and one with a specific parallel in panel evidence from 59 countries that similarly finds no sanitation–stunting association once region-level confounding is netted out. A separate, preliminary extension asks, within its own three-country sample, whether modelled local WASH coverage during a child’s specific early-life postnatal window is associated with later anthropometric outcomes — a measurement response to one dimension of the single-round baseline design’s inability to assign exposure near each child’s own developmental period, not an attempt to compare baseline and historical results as a test of exposure timing, since the two samples differ in several other respects (survey rounds, exposure construction, eligibility rules, and control sets) that this design cannot separate from timing. This extension’s pooled, established-date water coverage–height-for-age association is positive and statistically significant ($\beta = 1.03$ per 10 percentage points of modelled coverage, 95% CI [0.51, 1.54], $p < .001$), independently reconstructed and numerically confirmed with a separate implementation and sample during this project, but it is reported alongside substantial, explicitly documented methodological qualifications: sensitivity to the pooled weighting convention, an undocumented measurement-subsample sampling weight, and area- rather than population-weighted exposure averaging around confidentiality-displaced survey coordinates. All results in this thesis are associational, estimated under a cross-sectional design, and are not interpreted as causal effects or as evidence either for or against the effectiveness of WASH investment as a policy instrument. The thesis’s contribution is a directly comparable household-versus-community WASH analysis within one four-country framework, together with a carefully bounded first attempt at geospatially matched early-life exposure, each reported with its own methodological limitations treated as integral to its interpretation.

1. Introduction

Child stunting remains one of the most persistent markers of population health disadvantage in Sub-Saharan Africa. Linear growth faltering concentrates in the first two years of life and is difficult to reverse once established, with consequences that extend well beyond anthropometry into cognitive development, school attainment, and adult earning capacity (Victora et al.; Victora et al.; Black et al.). Among the environmental exposures proposed to explain why children in similar economic circumstances grow at such different rates, water, sanitation, and hygiene (WASH) conditions occupy a central place in both the public-health and development-economics literatures. The proposed mechanism is biologically direct: inadequate water and sanitation increase a child’s exposure to faecal pathogens, raising the frequency of diarrhoeal and sub-clinical enteric infection, which in turn diverts nutritional intake toward immune response rather than growth — a pathway often described as environmental enteric dysfunction (Cumming and Cairncross). This biological mechanism is well established and is not disputed even by the strongest recent causal evidence against basic WASH interventions reducing stunting (Section 2.3 discusses this distinction precisely). What is considerably less settled is whether the mechanism translates into a *detectable, policy-relevant statistical association* in observational survey data, once household wealth and local development are accounted for, and whether a *basic facility-level* WASH classification — as opposed to a comprehensive intervention package — is the right exposure to look for it in. This thesis is concerned with that narrower, empirical question.

1.1 Two distinct exposures: a household’s own facilities and its surroundings

A household’s own water source and sanitation facility are not the only WASH-related exposure a child faces. Even a child in a household with an improved latrine and piped water can be exposed to pathogens through shared water sources, open defecation by neighbours, flooding, or simply proximity to inadequately managed waste if the surrounding community lacks similar facilities. This distinction — between **own-household WASH** as a private good and **surrounding-community WASH** as a local environmental condition — is not a modelling convenience; it corresponds to two different exposure pathways, and recent community-level WASH research (Donohue et al.; Skoufias and Vinha) suggests the two need not move together or carry the same association with child growth. This thesis constructs and compares both: a child’s own household classification, and a leave-one-out measure of improved-facility coverage among every *other* household in the same DHS cluster, so that a child’s own household is never used to construct its own community exposure.

1.2 Exposure timing in cross-sectional survey designs

A specific measurement limitation affects the cross-sectional, repeated-survey-round observational literature this thesis’s baseline analysis belongs to, including DHS-based studies generally (Rakotomanana et al.; Addae et al.) and this thesis’s own baseline: household and community WASH status is recorded once, at the survey date, and applied to every child in that round regardless of the child’s age. A four-year-old measured in a 2022 survey experienced their nutritionally critical first two years of life around 2018–2019, under WASH conditions

the survey-date measurement cannot describe, while an infant measured in the same round has WASH recorded very close to their own early life. This limitation is specific to designs of this kind; it does not describe the major WASH-intervention trials, which assign exposure near each child's own birth and follow growth prospectively over a defined window (Section 2.3), nor does it describe panel designs that track WASH and health indicators over successive rounds within the same geographic units (Section 2.2). Age controls and age-interaction tests, which this thesis also estimates, describe how the *current* survey-date association varies with how old a child is today; they do not reconstruct what a particular child's own environment actually was during their own developmental window. This thesis's main, baseline analysis documents this limitation explicitly and does not claim to resolve it.

A separate, clearly bounded extension responds to one specific dimension of this limitation: rather than asking whether assigning a *historical* exposure instead of a *current* one changes an otherwise-identical estimate — which this thesis's baseline and historical samples differ too much in other respects to isolate (Section 6.2 states this precisely) — the extension asks a self-contained question within its own sample: **is modelled local WASH coverage during a child's own early-life window associated with that child's later anthropometric outcomes?** It does so by matching each child's birth-cohort location to a multi-year, modelled local WASH coverage surface (IHME's gridded 2000–2017 drinking-water and sanitation access estimates (Local Burden of Disease WaSH Collaborators)) for a defined early-life window, which gives genuine within-cluster, cross-birth-cohort variation that a single survey round cannot offer. This is a measurement improvement on one specific dimension — assigning an exposure closer to each child's own developmental period, rather than to the survey date — not a resolution of the exposure-timing problem in general, and not a reconstruction of what any individual household's actual historical WASH facilities were, which no available data source for this thesis can provide. The extension's own, separate limitations (undocumented measurement-subsample weighting, DHS/IHME input overlap, residence used as a proxy for historical residence, confidentiality-driven coordinate displacement) are stated in full in Section 6 and govern how much weight its results can bear.

1.3 Research questions

This thesis asks three related questions. First, is a child's own household's water and sanitation status associated with their anthropometric outcomes (height-for-age, weight-for-age, weight-for-height), after accounting for household and cluster-level confounding? Second, is the WASH status of the surrounding community — independent of the child's own household — separately associated with these outcomes? Third, within the historical sample's own three countries, is modelled local WASH coverage during a child's specific early-life window associated with these outcomes? All three questions are posed and answered as **associations under a cross-sectional design**, not as estimates of causal effects; this is stated once here and governs how every coefficient in this thesis should be read. The third question is answered only within the historical sample; it is not compared to the first two questions' answers as if the comparison isolated the effect of exposure timing, for the reasons given in Section 1.2 and developed fully in Section 6.2.

1.4 Scope: four countries, two time horizons

The baseline analysis pools the most recent Demographic and Health Survey (DHS) round available, as of the data-collection date, for four countries: Ethiopia (2024–25), Ghana (2022), Kenya (2022), and Nigeria (2024). These four countries were selected for data availability and recency, not as a representative sample of Sub-Saharan Africa, and no claim in this thesis generalises findings from these four countries to the region as a whole. The historical extension uses an earlier DHS round for three of the four countries — Ghana (2014), Kenya (2014), and Nigeria (2018) — linked to IHME’s annual gridded WASH-coverage estimates, which are available through 2017. Ethiopia’s 2016 round is retained only as a separately labelled, provisional sensitivity analysis, because the Ethiopian-calendar interview-date conversion required to place Ethiopian children in a specific early-life calendar window could not be fully validated (Section 3.5 and Appendix C). The baseline and historical samples are **not the same children**, use different exposure definitions and spatial/temporal constructions, and are analysed, validated, and reported separately throughout; no result in one sample is used to adjust, select, or validate a result in the other.

1.5 Contribution

This thesis makes two separate contributions, in two separate samples, which should not be read as one combined design. In the **baseline sample**, the contribution is the explicit, side-by-side construction and comparison of own-household and leave-one-out community WASH exposure within one four-country sample and one consistent set of controls, each exposure estimated under the fixed-effects structure its own construction requires (Section 4.2–4.3). In the **historical sample**, the contribution is a transparently documented, preliminary attempt to introduce genuine early-life temporal exposure variation through geospatially matched historical coverage data; this extension examines a single modelled local-coverage exposure assigned to each child’s own early-life window (Section 3.5–3.8) and does **not** reproduce the household-versus-community comparison — the historical data source does not distinguish a child’s own household from its surrounding community, so no household/community contrast exists to draw within the historical sample. Having a particular combination of four countries is not, by itself, evidence of substantive novelty, and this thesis does not claim it as such; the specific design choices above are each reported with the methodological limitations of their own sample treated as integral to their interpretation rather than as closing caveats.

1.6 A restrained summary of findings

Stated briefly here and developed with full uncertainty in Sections 5–6: household-level WASH shows no statistically significant association with height-for-age once cluster fixed effects are used to compare households within the same community (water: $\beta = -0.085$, 95% CI $[-0.175, 0.004]$, $p = .062$; sanitation: $\beta = 0.026$, 95% CI $[-0.058, 0.110]$, $p = .548$). Community-level WASH coverage shows a large, strongly significant unadjusted association that attenuates to statistical insignificance once socioeconomic controls and region fixed effects are added (water: 0.447 unadjusted vs. 0.033 adjusted, $p = .573$; sanitation: 0.472 unadjusted vs. 0.008 adjusted, $p = .894$) — a pattern consistent with, but not conclusive proof of, confounding by local so-

cioeconomic conditions. Within the historical sample, the pooled established-date water–HAZ association is positive and significant (β per 10 percentage points of modelled coverage = 1.03, 95% CI [0.51, 1.54], $p < .001$) but is sensitive to the weighting convention used (falling to 0.41 unweighted on the same observations) and carries a substantial list of unresolved documentation issues (Section 6; Appendix C). None of these results should be read as evidence either for or against the effectiveness of WASH investment as a policy instrument; they describe statistical associations in specific survey samples, nothing more.

1.7 Design and principal limitations, stated at the outset

This is a cross-sectional, associational design throughout. It is not a panel, not a difference-in-differences design, and not an instrumental-variables design, and none of those terms is used elsewhere in this thesis to describe a method that is not actually implemented. The principal limitations — unresolved sampling-weight documentation for the historical extension’s measured-anthropometry subsample, DHS/IHME input overlap in two of the three historical-extension countries, the use of current residence as a proxy for historical residence, confidentiality-driven displacement of DHS cluster coordinates, and the small number of countries in both samples — are stated in full in Section 6 and are not treated there as a closing formality.

1.8 Structure

Section 2 situates this thesis within the WASH–child-growth literature, distinguishing biological plausibility from the specific question of whether a facility-level classification shows a detectable association, and distinguishing causal evidence from conditional association. Section 3 describes the data, sample construction, and variable definitions for both the baseline and historical samples. Section 4 sets out the empirical strategy, with notation matched to the actual estimating equations. Section 5 reports results. Section 6 discusses findings in relation to the literature, states in detail why the baseline and historical results are not directly comparable as a test of exposure timing, and states all limitations in full. Section 7 concludes. A note on the thesis title, and other administrative items not yet confirmed, are recorded in the author checklist accompanying this manuscript package, not in the chapter text.

2. Literature review and conceptual framework

2.1 From biological plausibility to demonstrated association

The hypothesised pathway from WASH to child growth is biologically direct: inadequate water and sanitation increase exposure to faecal–oral pathogens, raising the frequency of diarrhoeal disease and sub-clinical environmental enteric dysfunction, which together redirect energy and nutrients away from linear growth (Cumming and Cairncross). This mechanism is not challenged even by the strongest recent evidence against basic WASH interventions reducing stunting: the authors of the three major WASH trials discussed below state explicitly that “the biological plausibility of WASH as public health interventions is not challenged by these findings” (Cumming et al., consensus message 2), while arguing that the *specific, basic* interventions they tested were insufficient to translate that plausible mechanism into a detectable growth effect. Black et al. (2013) place WASH-related infection among the proximate determinants of child undernutrition in their widely cited *Lancet* framework, alongside inadequate diet and maternal and child care practices, and Victora et al. (2010) document that growth faltering is concentrated in the window from birth to roughly 24 months. What this literature review is concerned with is a narrower, empirical question: does a *facility-type classification* — whether a household has an “improved” water source or sanitation facility, in the standard WHO/UNICEF JMP sense used throughout this thesis — show a detectable, policy-relevant statistical association with child growth once the obvious confounders are accounted for, as distinct from the question of whether WASH-related infection can plausibly affect growth at all.

2.2 Observational evidence and the confounding problem

A substantial observational literature finds WASH variables associated with child anthropometric outcomes in DHS and related survey data. Rakotomanana et al. (2020) link WHO/UNICEF JMP water and sanitation indicators to linear growth in children aged 6–23 months across several East African DHS surveys and find associations in the expected direction. Addae et al. (2024), using Ghana’s most recent national survey, find household sanitation and socioeconomic status jointly associated with both stunting and wasting among children aged 6–23 months, and unprotected drinking water associated with wasting but not stunting. Günther and Fink (2010) pool 172 DHS surveys and estimate that improved water and sanitation are associated with lower child mortality, and Spears (2013) documents that cross-country variation in average child height correlates with cross-country variation in open-defecation rates.

The central methodological difficulty common to this literature, and squarely faced in this thesis, is that household and community WASH status are not randomly assigned: both are strongly correlated with household wealth, parental education, and local infrastructure, all of which independently predict child nutrition through channels unrelated to pathogen exposure. Headey and Palloni (2019) address this directly, constructing a panel of 442 subnational regions across 59 countries from repeated Demographic and Health Surveys and estimating difference-in-differences regressions that exploit within-region change over time, net of region fixed effects, rather than cross-sectional variation alone. Their results are mixed across outcomes and should be read precisely rather than summarised as a single attenuation story: improved water access is statistically insignificantly associated with most outcomes, although water piped di-

rectly into the home specifically predicts reductions in child stunting; sanitation improvements predict sizeable reductions in diarrhoea prevalence and child mortality but are *not* associated with changes in stunting or wasting once the within-region panel design is used. This last finding — a null sanitation–stunting association once region-level time-invariant confounding is netted out — is a closer and more precise parallel to this thesis’s own null household- and community-sanitation findings (Section 5) than a general “attenuation” story would suggest, and it is drawn on that way in Section 6 rather than as evidence that WASH associations with child health “generally” weaken under better controls, which overstates what this one study, on its own, establishes.

2.3 Credible causal evidence: what the major trials actually found, and what they did not test

Randomised evidence on WASH and child growth comes primarily from three cluster-randomised controlled trials, described precisely here, trial by trial and arm by arm, because loose paraphrase of their design risks both mischaracterising a genuinely different exposure structure from the cross-sectional literature in Section 2.2 and 2.5, and attributing a finding from one arm to a different intervention than the one that produced it. WASH-Benefits Bangladesh, WASH-Benefits Kenya, and the Sanitation Hygiene Infant Nutrition Efficacy (SHINE) trial in Zimbabwe each enrolled pregnant women and their children *in utero* and followed them prospectively for 18–24 months, assigning interventions near the start of that follow-up window, not at a single arbitrary survey date unrelated to each child’s own age. This is a materially different exposure-timing structure from the DHS-based observational studies discussed in Section 2.2 and 2.5, and the exposure-timing limitation described in Section 1.2 for this thesis’s own baseline design does not apply to these three trials.

Each trial used a factorial design with **separate arms**, not a single combined treatment: a water arm (point-of-use chlorination), a sanitation arm (an improved household or compound-level latrine), a hygiene arm (a handwashing station with soap), a combined-WASH arm (all three together), a nutrition arm (lipid-based nutrient supplementation and feeding-behaviour promotion, with no WASH component), and a combined WASH-plus-nutrition arm, each compared against a control. On linear growth specifically, **every individual WASH-related arm — water, sanitation, hygiene, and combined WASH — showed no statistically detectable effect in all three trials** (Cumming et al., pp. 2, 5 and Fig. 2). The nutrition arm, which contained no WASH component, showed a separate, modest positive effect on linear growth, described by the trial authors as a small gain despite high compliance (Cumming et al., p. 6); the combined WASH-plus-nutrition arm showed **no additional growth benefit over the nutrition arm alone** (Cumming et al., p. 2) — that is, adding WASH to nutrition did not improve on nutrition by itself. This is reported here, trial-component by trial-component, specifically so that the nutrition arm’s modest growth effect is not read as a WASH finding: it was produced by an arm with no WASH intervention at all. On diarrhoea, results were genuinely mixed across the WASH-related arms and sites — a substantial relative reduction in Bangladesh (against a low baseline prevalence), no detectable effect in Kenya, and no detectable effect in Zimbabwe (Cumming et al., pp. 2–5, Fig. 2).

“No statistically detectable effect” is the precise description used throughout this discussion, not “proven to be exactly zero”: these are adequately powered trials with high internal validity, and the authors themselves describe the growth result as consistent evidence across three sites,

but a null result in a hypothesis test establishes an absence of detected effect at the trial’s power and sample size, not mathematical certainty that the true effect is zero. The authors’ own interpretation, discussed next, is also explicit that this finding should not be generalised to *every possible* WASH intervention.

The consensus statement summarising these trials is explicit that this finding does not challenge the biological plausibility of WASH affecting growth (Section 2.1) and attributes the null growth result instead to the limited scope of the specific interventions tested: household-level, “basic” WASH services (in the JMP ladder sense) that left community-level open defecation and the broader water-supply infrastructure largely unchanged, rather than the “safely managed,” piped, networked services the authors argue may be needed for a detectable population-level effect (Cumming et al.). The authors explicitly call for research on more comprehensive, “transformative” WASH packages rather than concluding that WASH cannot matter for growth. This distinction matters directly for how this thesis’s own results should be read: the exposure measured throughout this thesis — household- or community-level “improved” facility status — is, like the trial interventions, a facility-type classification rather than a comprehensive intervention package, so a null or weak association in this thesis’s data is consistent with, though does not on its own confirm, the trial literature’s finding that facility-level WASH alone is often insufficient to move linear growth; it is equally consistent with confounding masking a real effect, with measurement error, or with this thesis’s own specific limitations (Section 6). This thesis does not interpret its null or weak findings as evidence that WASH investment in general is ineffective.

A related, quasi-experimental literature targets infant mortality rather than anthropometric growth, using a different identification strategy worth describing precisely. Geruso and Spears (2015) investigate why Muslim infants in India survive to their first birthday at a substantially higher rate than Hindu infants, despite Muslim households having lower wealth, consumption, and education on average — a puzzle documented by prior literature and not explained by household characteristics alone. The authors show that neighbourhoods with a higher share of Muslim households are, on most observable dimensions, *worse* places to raise a child, consistent with Muslims’ general socioeconomic disadvantage in India — with one specific, striking exception: open defecation. Hindus practise open defecation at a rate roughly 25 percentage points higher than Muslims nationally, a religious behavioural difference not explained by relative wealth, so neighbourhoods with more Muslim households have systematically better sanitation conditions even though they are worse on nearly every other observable predictor of child health. This lets the authors isolate a sanitation-specific channel from general neighbourhood quality, because the two move in *opposite* directions across religious composition rather than together, and they find this sanitation channel accounts for a substantial share of the Muslim mortality advantage. This thesis cites the working-paper version of this study specifically (version and verification details recorded in `bibliography_verification_ledger.md`, not restated here); it is cited for its identification strategy — exploiting a source of neighbourhood sanitation variation plausibly distinct from general local development — as an example of the kind of design this thesis’s own four-country, cross-sectional comparison cannot replicate, not as direct evidence about child growth, which Geruso and Spears do not study.

2.4 Household versus community exposure

Most of the DHS-based observational studies reviewed here measure WASH at the household level only. A smaller, more recent strand examines community-level or spatially aggregated

WASH exposure as a distinct construct. Donohue, Church, Assaf, and Mayala (2023), in a DHS Spatial Analysis Report, construct community-level improved-sanitation coverage (from a geospatial model, not a leave-one-out DHS-census rate) and relate it to childhood stunting in two countries, Nigeria and Zambia (both 2018 DHS), using multilevel logistic regression. The result is mixed, not a general finding, and is stated precisely here rather than paraphrased into one: in Nigeria, community coverage remained significantly associated with lower odds of stunting after full adjustment, though only modestly (adjusted OR = 0.97, 95% CI [0.95, 0.99] per 10-percentage-point increase), while household-level improved sanitation was no longer significant in the same adjusted model; in Zambia, community coverage was not significantly associated with stunting in either the unadjusted or adjusted model, and neither was household sanitation. This motivates constructing and reporting both household and community exposures as a worthwhile empirical question — the two measures can diverge — but it is not evidence that community coverage generally “carries explanatory relevance beyond” household facilities, which overstates a result significant in one of the two countries studied. Skoufias and Vinha (2026) develop a related community-externality framework for child health in Sub-Saharan Africa using DHS data from twenty countries, and their result is similarly specific rather than general: they find **no significant externality from community-level water coverage**, but do find community-level improved sanitation significantly associated with lower fever/diarrhoea incidence and higher height-for-age, with effects varying between urban and rural areas. This is a closer and more precise parallel to this thesis’s own baseline finding (Section 5.2) that community water and sanitation do not behave identically under full adjustment, though neither source’s result should be read as confirming this thesis’s specific coefficients, which come from a different sample, country set, and model. This thesis’s leave-one-out community construction follows this same logic; the author is not aware of a study pairing this exact leave-one-out construction with a household-level comparison in the same four-country sample, though this thesis does not claim to have conducted an exhaustive search of the literature sufficient to assert this combination is unprecedented more broadly — only that reviewing household and community WASH side by side, under the fixed-effects structure each exposure’s own construction requires, is the specific design choice this thesis makes and did not find replicated in the sources reviewed here. Balk et al. (2004), working on child mortality in West Africa with spatially referenced survey data, is an earlier methodological precedent for treating DHS-cluster-level geography as a unit of analysis distinct from the household, though applied to mortality rather than anthropometric outcomes.

A specific identification issue arises when community WASH is measured this way: because the leave-one-out rate is itself a property of the cluster rather than of the individual child, attempting to estimate it with cluster fixed effects — the natural approach for household WASH — would remove nearly all of its substantive variation, leaving a coefficient identified almost entirely by the small mechanical residual that arises because a different household is excluded from each child’s own calculation (Section 4 derives this precisely). This thesis therefore uses region, not cluster, fixed effects for the community specification.

2.5 Exposure timing in single-round, cross-sectional survey designs specifically

The limitation introduced in Section 1.2 applies to studies that measure WASH status once, at a single survey date, and relate it to the anthropometric status of children of varying current ages within that same round — a description that fits this thesis’s own baseline design and the DHS cross-sectional studies reviewed in Section 2.2 (Rakotomanana et al.; Addae et al.;

Günther and Fink; Spears), but does **not** describe the major WASH trials (Section 2.3), which assign and measure exposure prospectively from near each child’s own birth, nor Headey and Palloni’s (2019) panel design, which exploits change over successive survey rounds within the same subnational regions rather than a single round’s cross-section. Age controls and age-interaction terms — which this thesis also estimates for its baseline sample (Section 4, Section 5) — describe whether the *current* survey-date association differs by how old a child is today; they do not reconstruct what a child’s own environment was during their own early life, and this thesis is explicit that these are different empirical questions. This single-round limitation, specific to this design class, is the reason this thesis’s historical extension attempts a different, early-life-window exposure assignment within its own separate sample (Section 1.2), not a reason to expect the two samples’ results to isolate an exposure-timing effect when compared to each other (Section 6.2).

2.6 Geospatial exposure assignment, and a matching-method precedent from outside the WASH literature

Assigning a continuous, gridded exposure surface to a survey respondent’s location requires handling two well-documented DHS features. First, coordinates are randomly displaced for confidentiality (Burgert et al.; Pérez-Heydrich et al.), which affects any extraction method: whichever pixel or pixels are read, they are read around the cluster’s *recorded*, displaced coordinates, not its true location, and no extraction method recovers the true location or corrects for the (undisclosed, per-cluster) displacement actually applied. Second, a value still has to be read from the surface at that location, and more than one method exists for doing so — a containing-pixel extraction, which reads the single pixel the point falls in, or a buffer-based extraction, which averages pixels within some radius, by simple area weighting or by population weighting if population data for the area are available. This thesis uses an area-weighted buffer average as its primary method and a containing-pixel extraction as a labelled sensitivity (Section 3.6; Section 5.8); the two are not interchangeable descriptions of the same procedure, and the choice between them is made explicit here rather than treated as if only one were possible. Johnson, Jacob, and Brown (2013) provide an earlier precedent for linking satellite-derived environmental surfaces (forest cover) to DHS cluster locations to study a different child-health outcome. Blom, Ortiz-Bobea, and Hoddinott (2022) match historical, geocoded weather data to each child’s own developmental window across repeated DHS rounds — a design this thesis’s historical extension borrows structurally. This source was reviewed at the level of its published abstract and design description, not independently verified claim-by-claim in this pass (recorded in `bibliography_verification_ledger.md`); it is cited here, and only here, for its general matching-method structure — matching a time-varying, geocoded exposure to each child’s own developmental window rather than to the survey date — not for any specific numerical finding, and not as evidence about WASH or nutrition, which that study does not examine.

2.7 This thesis’s contribution and the gap it addresses

This thesis combines three design elements that this literature review has not found combined in one study: household and leave-one-out community WASH compared directly within the same sample and controls (Section 2.4); and, in a separate, bounded, and explicitly preliminary ex-

tension, an attempt to assign modelled local WASH coverage to each child’s own early-life window using geospatial matching (Section 2.6), within the historical sample’s own three countries. That a specific four-country combination has not been studied before is not, on its own, evidence of a substantive methodological contribution, and this thesis does not rely on country selection as its claim to contribution. The methodological contribution claimed is narrower: a direct, consistently controlled comparison of the two exposure levels, and a transparent, fully documented attempt — including its own unresolved issues — at an early-life exposure-matching design that neither Donohue et al. (2023) nor Skoufias and Vinha (2026) implement; both examine current-round community WASH exposure (geospatially modelled or DHS-census-based respectively), not a time-varying exposure matched to each child’s own developmental window, and neither source is cited here as calling for that specific extension — the gap is this thesis’s own observation, not an attributed recommendation. The gap this leaves open, and does not claim to close, is the same one Headey and Palloni (2019) and the trial literature (Cumming et al.) already identify at a larger scale: credibly separating a WASH-specific effect from the broader package of local development improved facilities tend to accompany. This thesis’s historical extension narrows the exposure-timing dimension of that problem within its own sample; it does not resolve the confounding dimension, and it does not resolve the exposure-timing dimension for the baseline sample, since the two samples differ in several other respects simultaneously (Section 6.2).

3. Data and variable construction

This section documents the baseline and historical samples separately throughout. They are built from different DHS rounds, use different exposure constructions, and are validated independently; no claim in this section about one sample should be read as applying to the other. Numerical claims are sourced to the authoritative files listed in Appendix A; this section states what each variable is and how it was constructed, not the estimation results themselves (Section 5).

3.1 Baseline sample: unit of analysis, rounds, and pooling

The unit of analysis throughout the baseline sample is the child. The baseline pools the most recent DHS round available, as of data collection, for four countries: Ethiopia (2024–25), Ghana (2022), Kenya (2022), and Nigeria (2024). The full pooled, harmonised child file contains 70,231 children across 4,481 DHS clusters and 114 administrative-region fixed-effect cells (one set of regions per country, pooled). This pooled file is **not** the regression sample for any model: it is the universe before outcome-validity and exposure-classifiability restrictions are applied.

3.2 Baseline anthropometric outcomes and under-five eligibility

Three outcomes are constructed following standard DHS/WHO anthropometric practice: height-for-age (HAZ), weight-for-age (WAZ), and weight-for-height (WHZ), each a standard-deviation z-score relative to the WHO Child Growth Standards reference population (WHO Multicentre Growth Reference Study Group et al.). A child contributes to an outcome only if that outcome’s z-score is flagged valid in the DHS recode (implausible values, per DHS’s own flagging convention, are excluded) and the child falls within the DHS-defined under-five anthropometric-measurement universe. Because valid measurements are not available for every child in the pooled file, the outcome-specific baseline samples are smaller than the full pooled file: 36,985 children (HAZ), 37,260 (WAZ), and 37,088 (WHZ). Comparing the household and community specifications on identical observations further requires both exposures to be classifiable for every child in the comparison, giving common baseline samples of 36,040 (HAZ), 36,308 (WAZ), and 36,145 (WHZ), spread across 4,426 DHS clusters. Under-five age is taken from the DHS-reported completed age in months where delivered directly by the recode (Ethiopia, Nigeria), and from a documented calendar-month difference construction otherwise (Ghana, Kenya), following each country round’s own recode conventions.

3.3 Baseline WASH exposures: household and leave-one-out community

Four exposures are constructed for the baseline sample: own-household improved/non-improved water, own-household improved/non-improved sanitation, and their leave-one-out community counterparts, each following the WHO/UNICEF Joint Monitoring Programme (JMP) “improved” facility classification (piped water, protected wells/springs, and similar for water; sewer/septic and improved latrines for sanitation — the same classification independently verified for the

historical extension’s gridded product in Section 3.4 and Appendix A). Household WASH is a household-level binary indicator. Community WASH is constructed for each child from the DHS household census of *every other household in the same DHS cluster, excluding the child’s own household* — a leave-one-out construction, not an average over the other children in the analytical sample, so that two children in the same household never contribute to their own household’s community measure and a child’s own classification is never part of its own community exposure. This construction makes community WASH, by design, a property of the cluster that varies mainly *between* clusters rather than within them; Section 4.3 derives precisely how little within-cluster variation the leave-one-out construction leaves and why this rules out a cluster-fixed-effects specification for the community exposure specifically, while it remains appropriate for the household exposure.

3.4 Baseline covariates, geography, and weights

Household- and child-level covariates include child sex, maternal age, maternal education (categorical), literacy, household size, birth order, wealth quintile, and urban/rural residence. DHS cluster (IDHSPSU, sample-qualified so that a cluster code never spans two survey rounds) and administrative region are retained as geographic identifiers; DHS cluster GPS coordinates are randomly displaced for confidentiality (urban clusters up to 2 km; rural clusters 99% up to 5 km and 1% up to 10 km; (Burgert et al.)), which is not consequential for the baseline models (which do not use coordinates directly) but is directly consequential for the historical extension’s spatial matching (Section 3.5). Baseline models use PERWEIGHT, DHS’s standard child-level sampling weight, normalised within each model’s own estimation sample so that each country contributes an equal total weight mass regardless of its raw sample size — an authorised pooling choice, not a claim that any single country’s raw sample size should determine its influence on a pooled coefficient (Section 4.5 gives the exact formula and its consequences).

3.5 Historical extension: rounds, exposure source, and scope

The historical extension links an earlier DHS round for three countries — Ghana (2014), Kenya (2014), and Nigeria (2018) — to IHME’s annual gridded estimates of local access to improved drinking water (W_IMP) and improved sanitation (S_IMP), available for 2000–2017 (Local Burden of Disease WaSH Collaborators). These product definitions were verified directly from the primary source during this project’s development, not inferred from the 0–100 raster scale or file names alone: W_IMP is the sum of piped water (on- or off-premises) and other JMP-improved sources (protected wells and springs, bottled water, rainwater collection, bought water); S_IMP is the sum of sewer/septic sanitation and other JMP-improved facilities (improved and ventilated improved latrines, composting toilets), each reported by the source as percent of population with access on a continuous ~5×5 km surface. This project’s own spatial extraction computes an **area average of the raster surface within each child’s exposure buffer**, not the source study’s own population-weighted administrative aggregate; this distinction is maintained throughout and restated wherever a historical coefficient is reported (Section 5.3; Section 6). Ethiopia’s 2016 round is included only as a separately labelled, provisional sensitivity analysis (Appendix C), not as part of the established-date pooled sample, because the Ethiopian-calendar-to-Gregorian interview-date conversion required to place Ethiopian chil-

dren’s exposure windows could not be validated to the same standard as the three established-date rounds (Appendix C details the three candidate conversions and why one of them, labelled C1, is invalid and excluded from reporting). As stated in Section 1.2, this construction assigns each child a *modelled, area-level* local coverage estimate for a defined early-life window; it is a measurement improvement on the timing dimension specifically, not a record of what that child’s own household facilities actually were, which this or any other available data source cannot provide.

3.6 Historical spatial extraction: buffers, displacement, and coverage

Each child’s exposure buffer is centred on its DHS cluster’s (displaced) coordinates using an azimuthal equidistant projection local to that cluster, which avoids the area-distortion that a single fixed projection introduces at different latitudes. Within that buffer, the raster’s value is averaged using an exact disk–pixel overlap computation (not a centroid or nearest-neighbour approximation), so that a pixel partially inside the buffer contributes its exact area-weighted share. Coverage completeness is checked explicitly: a child-year is assigned a valid extraction only when the buffer’s valid-pixel area meets a documented tolerance (within 10^{-3} of the full buffer area), and partial- or missing-coverage child-years are excluded from that window rather than silently assigned a value computed from incomplete coverage.

3.7 Historical exposure windows and completion rules

Four candidate exposure windows are constructed for each child, applied to the same underlying annual raster series: the calendar year of birth; the nine months before birth (prenatal); the first 12 completed postnatal months (the working main window; Section 4.1 states why); and the first 24 postnatal months. Because IHME’s product is an annual surface, a window spanning two calendar years is assigned a weighted average of the relevant years’ values, weighted by how many of the window’s months fall in each calendar year. A child contributes to a given window only if every calendar year the window touches has a verified raster value **and** the window has been completed by the child’s interview date (so a child interviewed before their own first-year postnatal window has elapsed is excluded from that window, not assigned a partial-year value). This completion rule means that, even for windows whose exposure *value* does not depend on the precise interview date (Appendix C shows this is true for the birth-year window), eligibility for inclusion still requires a valid interview-date conversion — relevant to why Ethiopia’s three interview-date candidates matter even for windows that are otherwise interview-date-insensitive.

3.8 Historical covariates, weights, and the established-date measurement universe

Historical models control for cluster fixed effects, survey-qualified country-by-birth-year terms (each country’s own birth-year effects relative to that country’s own earliest observed birth year, not a single pooled reference — Section 4.5 explains why this distinction matters for the pooled estimate), child sex, maternal age, maternal education, and marital status, with the child’s own completed age in months (reported where delivered, documented calendar-month construction

otherwise, matching the baseline’s rule in Section 3.2) entered as an additional linear control. Household wealth quintile is **excluded from the historical extension’s main specification**, for a documented, non-fixed-effects reason: current wealth may be a mediator or a post-exposure outcome of the historical WASH exposure being studied, and wealth quintile is correlated with the WASH exposure itself, so including it in the main specification risks conditioning on a variable downstream of the exposure of interest; wealth is included only as a labelled sensitivity check (Section 5.3), where the pooled water–HAZ coefficient is essentially unchanged (1.01 vs. 1.03). This is a different, and more defensible, reason than “cluster fixed effects absorb wealth” — that claim is false in general, since cluster fixed effects absorb only what is constant *within* a cluster, and wealth varies across households within the same cluster just as WASH status does (confirmed directly for this thesis’s own baseline cluster-fixed-effects model, where wealth quintile *is* included and estimated as a standard covariate; Section 4.4).

Weighting uses PERWEIGHT, normalised so that each of the three established-date countries contributes an equal total observation weight within each model’s own final estimation sample (not population-size weighting) — the same authorised convention as the baseline, carried over for comparability, and reported alongside its unweighted counterpart in every case (Section 5.3). The documented sampling weight for measured anthropometry specifically — as distinct from the general household/individual sampling weight — could not be located in official DHS documentation despite repeated attempts at alternative official sources (Appendix A records the exact sources tried and their outcomes); this project’s working inference, stated explicitly wherever historical coefficients are reported, is that within-survey PERWEIGHT normalisation removes a constant measurement-subsample selection factor under an assumption of random household-level selection into the anthropometry subsample, but it cannot and does not correct for any differential non-response among eligible children who were not measured. Historical results are therefore reported as **weighted conditional associations among eligible measured children**, not as population-representative estimates for any defined population, and this phrasing is used consistently rather than varied across sections.

3.9 A documented measurement-universe discrepancy: Nigeria 2018

Three distinct child populations arise for Nigeria 2018, and this thesis does not force them to a common count, nor treat any apparent gap between them as resolved. The official 2018 NDHS final report counts 12,806 (unweighted) children under five as eligible for measurement in the survey’s anthropometry subsample (households selected for the men’s questionnaire, roughly one-third of all households) and actually measured. The IPUMS-DHS birth-history extract used in this thesis contains 33,924 birth-history records for children born within approximately five years of the survey, of which 11,704 are measured or flagged non-missing for anthropometry. The final historical-extension estimation sample (water, HAZ, the main post-12-month window) retains 6,912 children after applying the window-completion, spatial-coverage, and singleton-cluster exclusions described above. These three counts differ by construction — the official count is eligible children in subsample households; the extract is birth-history children of interviewed women, which can omit children whose mothers are not themselves in the sampled frame (a hypothesis raised in this project’s own documentation, not independently verified); and the estimation sample further requires complete exposure windows and complete spatial coverage. The gap between the official count and the extract (12,806 vs. 11,704 non-missing) is disclosed here as an open, unresolved measurement-universe question and is not treated as resolved by any estimation choice in this thesis.

4. Empirical strategy

Throughout, i indexes a child, h the child’s household, c the child’s DHS cluster, r the child’s administrative region (each cluster belongs to exactly one region), and s a survey round (country). All specifications below are estimated and reported in this thesis as **associational**, not causal; none should be read, described, or cited elsewhere in this thesis as identifying a causal effect. None of the designs below is a panel, a difference-in-differences design, or an instrumental-variables design — these terms are not used anywhere in this thesis to describe a specification that is not what is actually estimated.

4.1 Baseline: household WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta \text{WASH}_{ih} + X_i' \gamma + \delta_r + \varepsilon_{ihcr}$$

WASH_{ih} is the child’s own household water or sanitation status; X_i is the covariate set described in Section 3.4; δ_r is a fixed effect for each of the 114 pooled administrative regions. β compares children whose households differ in WASH status, using variation both between and within regions, net of the controls and of anything constant across an entire region. A region can contain many DHS clusters, so δ_r leaves room for clusters within the same region to still differ in local infrastructure, environment, or health-service access — characteristics plausibly correlated with both household WASH and child nutrition. This motivates the cluster-fixed-effects specification below.

4.2 Baseline: household WASH, DHS cluster fixed effects (the main household specification)

$$Y_{ihc} = \alpha_c + \beta \text{WASH}_{ih} + X_i' \gamma + \varepsilon_{ihc}$$

α_c is a separate fixed effect for each DHS cluster. β is now identified from differences in household WASH status *among households within the same cluster*, conditional on the covariates; α_c absorbs anything shared by every household in a cluster simply by virtue of being in that cluster — local infrastructure, environment, market access — that a region fixed effect could not distinguish between clusters. It does not absorb differences *between* households within the same cluster: two neighbouring households can still differ in wealth, education, or unobserved behaviour in ways connected to both their own WASH status and their children’s growth, which is why this specification remains associational rather than causal. One mechanical consequence: because every DHS cluster is entirely urban or entirely rural by DHS’s own sampling design, urban/rural residence is automatically absorbed by α_c and cannot be separately estimated in this specification — this reflects how DHS clusters are sampled, not a specification error. **Wealth quintile is included in X_i and is estimated, not omitted, in this specification;** cluster fixed effects absorb only what is constant within a cluster, and wealth varies across households within the same cluster (31.6% of clusters show within-cluster variation in own-water status and 50.6% in own-sanitation status — the variation that identifies β in the first place; an identically-constructed check confirms comparable within-cluster variation in wealth), so there is no mechanism by which cluster fixed effects would remove wealth from

this model, and none was observed to.

4.3 Baseline: community WASH, region fixed effects

$$Y_{ihcr} = \alpha + \beta \bar{C}_{-h,c} + X_i' \gamma + \delta_r + \varepsilon_{ihcr}$$

$\bar{C}_{-h,c}$ is the leave-one-out community coverage defined in Section 3.3. Because this exposure is, by construction, a property of the cluster rather than of the individual child, β here is identified mainly by comparing *clusters* with different community coverage within the same administrative region, net of the same controls. As with the household region-fixed-effects model, unobserved differences between clusters inside the same region are not removed, so this is an associational estimate of a contextual exposure, not an estimate of a causal spillover effect.

Why cluster fixed effects are not used for this exposure. A natural question is whether community WASH could also be estimated with cluster fixed effects, matching the household specification. It can be, but the resulting estimate would not be informative, for a specific, verifiable reason. The leave-one-out community rate is itself a cluster-level quantity: within a given cluster, every child's own community value differs from every other child's only because a different single household happens to be excluded from each child's own calculation — a mechanical artefact of the leave-one-out construction. Measured directly in this data, the within-cluster standard deviation of this rate is approximately 0.007 (water) and 0.011 (sanitation), against an overall standard deviation of approximately 0.35 for both. A cluster fixed effect would therefore absorb nearly all of the genuine, substantive, between-cluster variation in community coverage, leaving a coefficient identified almost entirely by this small mechanical residual rather than by real differences in community environments. This is a fundamentally different situation from the historical extension's cluster-fixed-effects specification (Section 4.6), where a real time dimension — birth-cohort variation within the same cluster — gives cluster fixed effects genuine, non-mechanical within-cluster variation to work with. Cluster fixed effects are therefore appropriate for household WASH and for historical WASH, but not for current-round community WASH, and this is a design consequence of the leave-one-out construction itself, not an inconsistency in this thesis's approach.

4.4 Baseline: household and community WASH jointly, and current-age heterogeneity

$$Y_{ihcr} = \alpha + \beta_H \text{WASH}_{ih} + \beta_C \bar{C}_{-h,c} + X_i' \gamma + \delta_r + \varepsilon_{ihcr}$$

β_H is the household association conditional on surrounding community coverage, and β_C is the community association conditional on the household's own status. Because household and community water measures are fairly closely correlated, this specification is reported as a secondary decomposition, not a preferred model.

$$Y_i = \alpha + \beta \text{WASH}_i + \sum_a \theta_a \text{AgeGroup}_{ia} + \sum_a \lambda_a (\text{WASH}_i \times \text{AgeGroup}_{ia}) + X_i' \gamma + \text{FE} + \varepsilon_i$$

The λ_a terms test whether the association between *survey-date* WASH and nutritional outcomes differs across children’s *current* ages. This is a heterogeneity test on the existing cross-sectional design (Section 1.2), not a reconstruction of what WASH conditions an older child actually experienced during their own infancy, and it is kept conceptually and terminologically distinct from the historical extension’s exposure-timing design throughout this thesis.

4.5 Inference, weighting, and the pooled estimand

Weighting. For a model with estimation sample of size N pooled across K countries, each child’s weight is

$$w_i = \text{PERWEIGHT}_i \times \frac{N/K}{\sum_{j \in \text{country}(i)} \text{PERWEIGHT}_j}$$

computed within that model’s own final estimation sample (after all exclusions, including singleton-cluster removal — see below), so that each country’s weights sum to exactly N/K . **This equal-country-total weighting is an authorised, disclosed working choice, not a correction for a sampling design:** it targets the average country-level association with each country given equal total observation weight, not a population-size-weighted estimate of the three (or four) countries’ pooled population. It is important to be precise about what this equal weighting does and does not imply for the resulting pooled coefficient. Equal country weight *totals* do **not** make the pooled common-slope coefficient the arithmetic mean of the country-specific coefficients: in a weighted least-squares/fixed-effects regression, each observation’s influence on the estimated slope depends on its weight **and** on how much residual exposure variation it contributes after the fixed effects and controls are partialled out — a country with little residual within-cluster exposure variation contributes comparatively little information to the slope even at an equal weight total. This thesis’s own historical pooled water–HAZ estimate illustrates the point concretely: the simple unweighted average of the three established-date countries’ own coefficients is 0.58, while the actual pooled coefficient is 1.03 (Section 5.3) — precisely because Kenya’s residual exposure variation after controls (0.20 percentage points) is roughly a third of Nigeria’s (0.65) and under half of Ghana’s (0.46), so Kenya’s imprecise, negative country-specific estimate receives comparatively little weight in the pooled slope despite its equal weight total. Every pooled coefficient in this thesis is reported alongside its country-specific components and each country’s residual exposure variation, so a reader can see which countries actually drive a pooled result rather than inferring equal influence from the weighting description.

Estimator and inference. Household- and historical-model cluster fixed effects are implemented as weighted within-cluster demeaning, exactly equivalent to weighted least squares with a full set of cluster dummy variables. Standard errors use the cluster-robust CRV1 sandwich estimator, clustered on the same DHS-PSU grouping used for the fixed effect, with the small-sample correction $c = \frac{G}{G-1} \times \frac{N-1}{N-K}$, where G is the number of clusters and K counts only the non-fixed-effect regressors retained (the project’s adjudicated convention when the fixed-effect grouping and the clustering grouping are the same variable; this thesis’s own baseline household cluster-fixed-effects model’s primary p-values — water HAZ $p = .062$, sanitation HAZ $p = .548$ — use this convention; an alternative convention that additionally penalises degrees of freedom for the absorbed cluster dummies gives wider, still-available sensitivity p-values of .080 and .574 respectively, reported as such wherever cited, never silently substituted for the

primary values). Confidence intervals use the t -distribution with $G - 1$ degrees of freedom. Singleton clusters — those contributing only one observation to a given model’s final estimation sample, which carry no within-cluster information and cannot contribute to identifying a cluster-fixed-effects coefficient — are dropped before estimation and their count is reported with every model.

Rank and identification are computational properties, not causal ones. Every reported model’s design matrix is checked for full column rank after the fixed-effects transformation, and any column found to carry no within-group variation (fully absorbed) is dropped and named explicitly rather than silently producing an unstable coefficient. This is a statement that the stated regression equation is computationally well defined and that its coefficient is the one the equation describes — it is not a statement that the equation identifies a causal parameter, and this thesis does not conflate the two.

4.6 Historical extension: cluster fixed effects with cohort controls

$$y_{ic} = \alpha_c + \beta E_{ic}(w, p) + \lambda \text{age}_{ic} + \gamma' X_{ic} + \delta_{s,b_{ic}} + \varepsilon_{ic}$$

$E_{ic}(w, p)$ is the annual-weighted IHME mean (product p , window w) assigned to child i in cluster c , on the 0–100 percentage-point scale (Section 3.4–3.7); β is reported per 1 and per 10 percentage points. α_c is the cluster fixed effect; $\delta_{s,b}$ is a survey-specific (country-specific) birth-year term — each country’s own set of birth-year fixed effects, estimated relative to that country’s own earliest observed birth year rather than one pooled reference year across all three countries. This distinction matters mechanically: because every DHS cluster belongs to exactly one country, any purely country-level mean difference is already fully absorbed by α_c , so a single combined reference cell across countries would introduce columns that duplicate information the cluster fixed effect already provides — this was confirmed directly during this thesis’s independent verification of the pooled model (Appendix B), where an initial reconstruction using a single pooled reference cell for the birth-year terms produced a materially different coefficient (1.39 vs. the correct 1.03) until corrected to match the country-local-reference construction described here. X_{ic} includes child sex, maternal age, maternal education, and marital status (wealth excluded from the main specification for the reason given in Section 3.8, not because cluster fixed effects absorb it).

Why cluster fixed effects are appropriate here but not for current-round community WASH (Section 4.3). The historical exposure $E_{ic}(w, p)$ varies not only across clusters but, genuinely, across birth cohorts *within* the same cluster — a child born in 2014 and a child born in 2017 in the same cluster can have experienced different modelled local WASH coverage during their own respective early-life windows. This is real temporal variation, not a mechanical construction artefact of the kind that rules out cluster fixed effects for the current-round leave-one-out community measure; the cluster-fixed-effects specification is therefore appropriate for the historical exposure, conditional on there being enough genuine within-cluster, cross-cohort variation for the fixed effect not to absorb nearly all of it — quantified, per model, as the residual exposure standard deviation after all controls (Section 5.3), which ranges from 0.20 to 0.88 percentage points depending on country and product. This range is modest relative to the 10-percentage-point scale conventionally used to report the coefficient (Section 5.3 states the ratio explicitly), and this thesis does not interpret a 10-point coefficient as describing an exposure contrast that is typically observed within a cluster.

Weighting and inference for the historical extension follow the same formulas as Section

4.5, with $K = 3$ for the established-date pooled models (Ghana, Kenya, Nigeria). Model-specific samples are constructed independently for every window and product combination, and singleton clusters are dropped and counted exactly as in the baseline (Section 4.5).

5. Results

Every table in this section is reused directly from the authoritative, validated output files listed in Appendix A; no coefficient was computed for this chapter. Coefficients are reported with 95% confidence intervals and exact p-values throughout. Where two inference conventions exist for a model, both are labelled explicitly (Section 4.5); the primary convention is used in prose discussion unless the sensitivity convention is named.

5.1 Household WASH

Outcome	Water β [95% CI], p	Sanitation β [95% CI], p	N	Clusters
Region FE				
HAZ	−0.035 [−0.110, 0.040], .357	0.022 [−0.045, 0.089], .522	36,040	4,426
WAZ	0.021 [−0.038, 0.081], .487	0.045 [−0.010, 0.100], .109	36,308	4,426
WHZ	0.050 [−0.015, 0.116], .130	0.046 [−0.011, 0.103], .116	36,145	4,426
Cluster FE (main household specification; PRIMARY inference convention)				
HAZ	−0.085 [−0.175, 0.004], .062	0.026 [−0.058, 0.109], .548	36,040	4,426
WAZ	−0.029 [−0.100, 0.043], .435	0.052 [−0.015, 0.120], .130	36,308	4,426
WHZ	0.012 [−0.062, 0.085], .758	0.053 [−0.014, 0.119], .122	36,145	4,426

No household-level water or sanitation coefficient reaches conventional significance in the cluster-fixed-effects specification, which is the specification this thesis treats as the main household result (Section 4.2): it compares households within the same DHS cluster, net of the covariates, and is the specification least exposed to cluster-level confounding of the two. Non-significance here is read as an inconclusive association in this sample under this design, not as proof that no association exists — the confidence intervals remain wide enough (for example, water–HAZ spans −0.175 to +0.004) to be consistent with a range of true associations. Under the alternative, more conservative FULL_DUMMY_DF inference convention (Section 4.5), the household cluster-FE water–HAZ p-value is .080 and sanitation–HAZ is .574; both conventions lead to the same substantive conclusion for this pair of coefficients.

5.2 Community WASH: the adjustment path

Stage	Water β [95% CI], p	Sanitation β [95% CI], p
Minimally adjusted	0.447 [0.311, 0.584], <.001	0.472 [0.356, 0.587], <.001
+ child controls	0.395 [0.263, 0.527], <.001	0.452 [0.340, 0.565], <.001
+ socioeconomic status	0.127 [0.010, 0.244], .034	0.008 [−0.117, 0.133], .900
+ SES + region FE (fully adjusted)	0.033 [−0.083, 0.150], .573	0.008 [−0.113, 0.129], .894

(HAZ; N = 36,985, 4,434 clusters, all four stages.) The community water and sanitation

coefficients are large and highly significant before adjustment and attenuate sharply — losing significance entirely by the fully adjusted stage — as socioeconomic status and region fixed effects are added. **This attenuation pattern is consistent with confounding by local socioeconomic conditions, but it does not, on its own, establish confounding as the sole explanation:** the same pattern would also be produced by, for example, measurement structure changing as additional controls absorb variance, or by other specification features that happen to coincide with the added controls. The same four-stage pattern holds for WAZ and WHZ, with one difference worth stating precisely: the WHZ water association is already small and statistically insignificant even before adjustment ($0.033, p = .545$) and remains so throughout, unlike the HAZ and WAZ water associations, which start large and attenuate to insignificance. Across all three outcomes, the fully adjusted community specifications do not show a consistent association between improved WASH and child anthropometric outcomes, at either the household or the community level — a conclusion read here as “this sample does not provide clear evidence of an association once the obvious confounders are controlled,” not as “WASH has no true relationship with child growth,” a materially stronger claim this design cannot support.

5.3 Joint household/community model

Outcome	Household water	Household sanitation	Community water	Community sanitation
HAZ	−0.085 [−0.167, −0.003], .043	0.024 [−0.047, 0.096], .507	0.120 [−0.008, 0.248], .066	−0.018 [−0.148, 0.112], .784
WAZ	−0.015 [−0.085, 0.056], .677	0.039 [−0.020, 0.098], .191	0.081 [−0.023, 0.186], .121	0.022 [−0.095, 0.138], .706
WHZ	0.039 [−0.038, 0.117], .318	0.036 [−0.025, 0.097], .244	0.019 [−0.100, 0.138], .750	0.045 [−0.073, 0.163], .446

($N = 36,040$ common sample, HAZ row; comparable for WAZ/WHZ.) One cell in this table reaches conventional significance: household water conditional on community water, HAZ ($\beta = -0.085, p = .043$), in an unexpected (negative) direction. Given the correlation between household and community water measures, and that this is one significant cell among twelve estimated in this table alone, this result is reported as a single finding rather than treated as a central conclusion of this thesis; it is not interpreted as evidence that improved household water is harmful.

5.4 Specification sensitivity: Nigeria community sanitation

Specification	Sanitation–HAZ β	p
No controls	0.274	.004
+ child controls	0.296	.001

Specification	Sanitation–HAZ β	p
+ socioeconomic controls	–0.457	<.001
+ region FE	–0.184	.035
Household cluster FE (own-household sanitation)	–0.007	.919

The Nigeria community sanitation–HAZ coefficient changes sign and more than doubles in magnitude as controls are added sequentially, and a block-by-block audit finds that removing wealth quintile alone from the full control set moves the coefficient from –0.457 back to –0.196 — the single largest movement of any control block, though this demonstrates sensitivity to specification, not proof that wealth is the specific mechanism behind the reversal. The corresponding household-level (cluster-fixed-effects) coefficient for Nigeria is close to zero and not significant. This instability is reported as evidence that the community-level, region-fixed-effects estimate for Nigeria sanitation is **not stable across specifications** and should not be read as evidence that improved sanitation harms child growth in Nigeria.

5.5 Current-age heterogeneity

Across the twelve water/sanitation joint-interaction tests estimated (two exposures $\times$ three outcomes $\times$ two fixed-effects structures), **eleven of twelve reject the null of no heterogeneity** at the 5% level; the single exception is the water interaction for WHZ under region fixed effects ($p = .300$). This indicates that the household-WASH association with current anthropometric outcomes varies with a child’s current age in most of the specifications tested. As stated in Section 4.4, this is secondary evidence about the *current* survey-date association and does not identify a developmental mechanism or reconstruct any child’s actual early-life exposure — that question is addressed only by the historical extension below, under its own, separate limitations.

5.6 Historical extension: primary established-date models

All cells below are the primary, post-12-month-window, weighted models (established-date sample: Ghana 2014, Kenya 2014, Nigeria 2018). Coefficient is per 10 percentage points of IHME-modelled local coverage. Every cell carries the same methodological qualifications stated in Section 3.8 and discussed fully in Section 6 — none is promoted above the others for being significant.

Water (W_IMP) — coefficient [95% CI], p

Outcome	Ghana 2014	Kenya 2014	Nigeria 2018	Pooled established
HAZ	1.65 [0.34, 2.96], .014	–0.61 [–2.07, 0.86], .417	0.70 [0.15, 1.25], .013	1.03 [0.51, 1.54], <.001
WAZ	1.06 [–0.16, 2.28], .088	–0.22 [–1.30, 0.86], .690	0.45 [0.00, 0.90], .050	0.65 [0.20, 1.10], .005
WHZ	–0.00 [–1.16, 1.16], .997	–0.16 [–1.29, 0.96], .776	–0.05 [–0.48, 0.38], .809	–0.08 [–0.50, 0.35], .717

Sanitation (S_IMP) — coefficient [95% CI], *p*

Outcome	Ghana 2014	Kenya 2014	Nigeria 2018	Pooled established
HAZ	−2.48 [−4.91, −0.05], .046	−0.21 [−0.53, 0.11], .190	0.20 [−0.34, 0.75], .467	−0.10 [−0.37, 0.17], .483
WAZ	−0.08 [−2.04, 1.88], .935	−0.15 [−0.38, 0.09], .223	0.02 [−0.44, 0.48], .937	−0.07 [−0.29, 0.14], .511
WHZ	1.64 [−0.27, 3.56], .092	0.00 [−0.22, 0.22], .980	−0.17 [−0.59, 0.24], .407	−0.03 [−0.24, 0.17], .742

Sample sizes (N children / G clusters) — identical for the water and sanitation tables above, since both exposures are estimated on the same anthropometric sample per outcome/country; shown once here rather than repeated in every coefficient cell

Outcome	Ghana 2014	Kenya 2014	Nigeria 2018	Pooled established
HAZ	N1,940 / G353	N14,166 / G1,498	N6,912 / G1,263	N23,018 / G3,114
WAZ	N1,940 / G353	N14,166 / G1,498	N6,951 / G1,264	N23,057 / G3,115
WHZ	N1,940 / G353	N14,166 / G1,498	N6,942 / G1,264	N23,048 / G3,115

The pooled established-date water–HAZ coefficient (1.03, $p < .001$) was independently reconstructed and re-estimated with a different estimation library and a separately rebuilt sample, matching the figures above to six significant digits (Appendix B). This establishes that the reported number is what the documented specification, correctly implemented, actually produces. It is not a multiplicity correction: no formal adjustment for testing 24 cells (2 products $\times$ 3 outcomes $\times$ 4 sample groups) has been applied anywhere in this section, and computational reproduction of a single coefficient does not substitute for one or establish that this result is any less likely than its siblings to reflect chance given the number of cells examined. This coefficient’s reporting status relative to its sibling cells is therefore unchanged by the computational check — all 24 cells carry the same unresolved documentation issues (Section 3.8, Section 6) — and the check does not establish causality.

Weighted vs. unweighted, same observations. The pooled water–HAZ coefficient is sensitive to the equal-country weighting convention: 1.03 (weighted, equal country totals) vs. 0.41 [0.03, 0.80], $p = .035$ (unweighted, identical 23,018 observations). This is a disclosed, authorised sensitivity to a stated design choice (Section 4.5), not evidence that either weighting is more “correct”; both are reported together whenever this coefficient is cited.

Residual exposure variation and the 10-point scale. The pooled water–HAZ model’s residual within-cluster exposure standard deviation after all controls is 0.43 percentage points — meaning the conventional 10-percentage-point coefficient scale is roughly 23 times larger than the exposure contrast that actually identifies the model. The effect corresponding to one residual-exposure standard deviation is approximately 0.044 HAZ standard deviations. The 10-point coefficient should therefore be read as a reporting convention for comparability across studies, not as a description of a change that is typically observed within a cluster.

5.7 Alternative windows and common-sample comparison

These two sensitivity families (window choice, §5.7; wealth/spatial method/completion rule, §5.8) were estimated for both products and all three outcomes (96 and 48 models respectively, within the 291-model total); the pooled water–HAZ cells are shown below as the headline case to avoid repeating all 144 cells in the main text. The corresponding sanitation and WAZ/WHZ sensitivity cells are not more or less favourable to a WASH association than the water–HAZ cells shown — most show the same pattern of wide, zero-spanning confidence intervals as the primary sanitation and WAZ/WHZ results in Section 5.6 — and are reported in full in the public aggregate tables (docs/provenance/historical_wash_preliminary_estimates_v1/), not reproduced here solely because water–HAZ happened to be significant.

Window	Pooled water–HAZ β [95% CI]	Sample
Post-12-month (main)	1.03 [0.51, 1.54]	Own-eligible, N = 23,018
Birth-year	0.54 [0.12, 0.96]	Own-eligible
Prenatal (9 months)	0.64 [0.24, 1.03]	Own-eligible
Post-24-month	1.75 [0.81, 2.70]	Own-eligible
Post-12-month, common-sample	1.64 [0.68, 2.60]	All-window-eligible common sample, N = 16,419

The post-12-month estimate changes from 1.03 (its own eligible sample) to 1.64 when restricted to the smaller common sample eligible under all four windows — a **sample-composition effect** (children with complete 24-month windows differ systematically from the full post-12-month sample), not evidence favouring one window’s validity over another. No window was selected because of its coefficient’s significance; all four are reported together, as planned before any coefficient was examined (Appendix C).

5.8 Further sensitivity: wealth, spatial method, completion rule

Sensitivity	Pooled water–HAZ β [95% CI]
Main (post_12m, weighted)	1.03 [0.51, 1.54]
Wealth quintile added	1.01 [0.50, 1.52]
Containing-pixel assignment (vs. buffer average)	0.68 [0.26, 1.10]
Rural 10 km buffer (urban unchanged)	1.01 [0.42, 1.60]
Stricter completion rule (+1 month)	0.95 [0.43, 1.47]
Excluding Ghana (DHS/IHME input overlap)	0.74 [0.24, 1.24]

Adding wealth leaves the coefficient essentially unchanged (1.01 vs. 1.03), consistent with the mediator/overlap reasoning in Section 3.8 rather than a cluster-fixed-effects absorption mechanism. The spatial-method and completion-rule sensitivities move the coefficient by amounts smaller than, or comparable to, the gap between the main weighted and unweighted estimates, and are reported without privileging any one over the main specification.

5.9 Ethiopia: provisional, excluded from the established-date pooled sample

Ethiopia's three interview-date candidates are not treated as equally valid sensitivity evidence. Candidate C1 uses a 30-day Ethiopian-calendar arithmetic conversion already shown, independently of the WASH analysis, to be invalid for the delivered recoded date fields (Appendix C); it is excluded from this section and from any reporting table, and its past numerical agreement with the other two candidates is not treated as evidence of robustness. Candidates C2 and C3 remain provisional (neither independently validated) and agree closely with each other (water–HAZ: -0.39 both; sanitation–HAZ: -0.45 both, both CIs including zero), reported here, in Appendix C, only as an explicitly exploratory sensitivity, never pooled into or compared against the established-date headline results above.

6. Discussion and limitations

6.1 Relating the findings to the literature

This thesis’s baseline findings sit consistently alongside the existing literature, though the comparison is precise rather than general. Headey and Palloni (2019), using a subnational-region panel across 59 countries with difference-in-differences regressions that net out region-level, time-invariant confounding, find that sanitation improvements predict large reductions in diarrhoea and child mortality but are **not** associated with changes in stunting or wasting once this within-region design is used, while improved water access is statistically insignificantly associated with most outcomes (piped water specifically is an exception, predicting reduced stunting). The null sanitation–stunting finding in their panel design is a close, specific parallel to this thesis’s own null household- and community-sanitation associations with HAZ (Section 5.1–5.2): both studies find that a sanitation–stunting association that is detectable in less-controlled specifications is not detectable once a design better suited to netting out confounding is used, each in its own, different way (region-level panel fixed effects there; household- or cluster-level fixed effects here). This is a narrower and more defensible claim than saying WASH–growth associations “generally” attenuate under better controls, which overstates what any one study, including this thesis’s own, establishes on its own.

The non-significant household cluster-fixed-effects associations in this thesis (water–HAZ $p = .062$; sanitation–HAZ $p = .548$) are also broadly consistent with the strongest available randomised evidence: the water, sanitation, hygiene, and combined-WASH arms of the WASH-Benefits Bangladesh, WASH-Benefits Kenya, and SHINE trials each showed no statistically detectable effect on linear growth in any of the three sites, despite high measured compliance — while, in the same trials, the separate nutrition-only arm (with no WASH component) showed a modest positive effect (Section 2.3) (Cumming et al.). This consistency should not be overstated into confirmation, for a reason stated precisely in Section 2.3: this thesis’s exposure is a facility-type classification in cross-sectional survey data, not a randomised intervention with prospective follow-up from near each child’s birth, so the two literatures answer related but not identical questions. The one statistically significant household-level result in this thesis — household water conditional on community water, HAZ, negative and borderline significant (Section 5.3) — is a single cell among twelve estimated in that table and is not treated as evidence that improved household water is harmful to child growth; no mechanism in the literature reviewed in Section 2 supports that reading, and this thesis does not offer one.

The historical extension’s positive, significant pooled water–HAZ association (1.03 per 10 points, $p < .001$) is independently computationally verified (Section 5.6, Appendix B), but that check is not a multiplicity correction: no formal adjustment for the number of cells estimated across both the baseline and historical analyses has been applied, so this result’s chance of reflecting one significant draw among many tested cells is not something this thesis has quantified or ruled out. It would be tempting to read this coefficient as evidence that introducing a time dimension recovers an association the single-round baseline design cannot detect. Section 6.2 explains in detail why this thesis does not draw that conclusion.

6.2 Why the baseline and historical results cannot be compared as a test of exposure timing

This is stated as its own subsection because it is the single most tempting overinterpretation this thesis’s own structure invites, and because the historical extension’s own research question (Section 1.2–1.3) is deliberately framed to avoid it: the historical extension asks whether modelled early-life local WASH coverage is associated with outcomes *within its own sample*, not whether assigning a historical rather than a current exposure changes an otherwise-identical estimate.

The baseline and historical samples differ along at least four dimensions simultaneously, not one:

1. **Different children and survey rounds.** The baseline uses 2022–2025 DHS rounds across four countries; the historical extension uses 2014–2018 rounds across three (Ethiopia excluded; Section 1.4). No child appears in both samples.
2. **Different exposure definitions and construction.** The baseline’s household and community exposures are built from DHS household-census survey responses at the survey date (Section 3.3). The historical exposure is a modelled, area-averaged, annual gridded coverage estimate from an external source (IHME), matched to a specific postnatal window (Section 3.4–3.7). These are not the same quantity measured at two different times; they are two different kinds of measurement.
3. **Different eligibility and completion rules.** The historical sample additionally requires complete spatial coverage and a complete, interview-date-verified exposure window (Section 3.7), restrictions that have no baseline-sample counterpart.
4. **Different control sets and identifying variation.** The baseline’s main household specification uses cluster fixed effects with region-level socioeconomic controls varying by specification; the historical specification uses cluster fixed effects with country-by-birth-year terms and excludes wealth from the main specification for a reason specific to the historical design (Section 3.8, Section 4.6). The historical cluster-fixed-effects model is identified by genuine within-cluster, cross-birth-cohort variation (Section 4.6); the baseline’s community model is deliberately *not* estimated with cluster fixed effects, for a reason specific to its own leave-one-out construction (Section 4.3) that has no bearing on the historical design.

Because all four of these differ at once, a difference between the baseline’s largely null associations and the historical extension’s one significant association cannot be cleanly attributed to “the effect of assigning historical instead of current exposure.” It could instead, or in addition, reflect different confounding structures across the two country sets, different measurement error properties of a modelled surface versus a survey response, the different control sets, or chance, given the number of cells estimated in each sample (Section 5.6’s multiplicity point applies here too). This thesis does not attempt to adjudicate between these possibilities, because its design — two separately constructed samples, not a shared sample with only the exposure-timing dimension varied — cannot do so.

What the within-historical-sample window comparison does and does not hold constant. Section 5.7 compares four exposure windows (birth-year, prenatal, post-12-month, post-24-month) within the *same* historical sample’s country set and estimation approach. This comparison holds constant the children’s survey rounds, the exposure source and spatial-extraction method, the control set, and the estimator; it varies only which months of the IHME surface are averaged into each child’s exposure value. It does **not** hold constant the estimation sample across all four windows (each window has its own eligible sample, differing by which children have complete coverage for that specific window; Section 5.7), and it does not speak at all to

the baseline-versus-historical comparison discussed above, which this paragraph is careful to keep separate.

6.3 Limitations: unresolved documentation

Two issues in this thesis’s historical extension remain genuinely unresolved documentation gaps, not design choices or inherent limitations. First, no DHS or IPUMS source could be located that documents a measurement-subsample or anthropometry-specific sampling weight, despite repeated attempts at the DHS Biomarker Manual and alternative official sources (Appendix A records the exact sources tried); this thesis’s working inference — that within-survey PERWEIGHT normalisation removes a constant measurement-subsample selection factor under an assumption of random household-level selection — is stated as an inference, not a documented fact, and cannot correct for differential non-response among children who were eligible but not measured. Second, the gap between Nigeria 2018’s official measured-child count (12,806) and the IPUMS birth-history extract’s non-missing anthropometry count (11,704) remains unexplained. Consistent with both points, this thesis reports its historical coefficients as **weighted conditional associations among eligible measured children**, not as population-representative estimates for any defined population — and the use of an unweighted estimate alongside the weighted one (Section 5.6) does not resolve this concern, since the selection question is about who was measured at all, not about how measured children are weighted relative to each other.

6.4 Limitations: authorised design choices, disclosed and tested

Three features of this thesis’s design are deliberate, disclosed choices, not errors. First, the equal-country-total pooled weighting (Section 4.5) targets an average-country association with each established-date country given equal weight, not a population-size-weighted estimate; its consequence — the pooled water-HAZ coefficient moves from 1.03 (weighted) to 0.41 (unweighted, same observations) — is disclosed and reported alongside the main estimate in every instance. Second, the working 12-month postnatal exposure window was chosen during the design phase, before any model was estimated, for a documented reason (an early-life window whose completion can be checked against the interview date); it is not described anywhere in this thesis as “preregistered,” and the literature basis for this specific window length is modest — Cumming and Cairncross (2016) support early-life exposure generally and, if anything, lean toward a 24-month window — which is why all four candidate windows (Section 5.7) are reported together. Third, wealth quintile is excluded from the historical extension’s main specification because it may be a post-exposure mediator of historical WASH conditions and because it is correlated with the WASH exposure itself (Section 3.8), a substantive identification concern, not a claim that cluster fixed effects remove wealth mechanically, which this thesis’s own baseline model demonstrates is false by including and estimating wealth directly (Section 4.2).

6.5 Limitations: inherent, requiring disclosure rather than resolution

DHS/IHME input overlap affects two of the three established-date historical countries (Ghana and Kenya), whose own DHS surveys feed into the IHME gridded estimates used as the exposure; this thesis reports the overlap-exclusion sensitivity (0.74 for the Ghana-excluded pooled estimate; Section 5.8) alongside the main estimate rather than treating the overlap as removable. Current residence, as recorded at the survey date, is used as a proxy for historical residence during a child's early-life window; any child who migrated between their early-life window and the survey date is assigned the wrong location's exposure, under the primary buffer-average method and under the containing-pixel sensitivity alike. DHS cluster coordinates are randomly displaced for confidentiality, and this limitation holds regardless of extraction method: an area-weighted buffer average (this thesis's primary method) reads pixels around the displaced coordinates, not the true ones, and does not recover the true location, does not validate or correct for the (undisclosed, per-cluster) displacement distribution DHS actually applies, and is not a population-weighted estimate (Section 3.4, Section 3.6); the containing-pixel sensitivity (Section 5.8) reads a single displaced pixel instead, which is subject to the same location uncertainty by a different mechanism — neither method is immune to it, and neither resolves it. This thesis's own spatial extraction, under its primary method, computes an area-weighted average of the raster surface within each buffer, which is a different quantity from the source study's own population-weighted administrative aggregate (Section 3.4). IHME's exposure surface is itself a *modelled* estimate with its own uncertainty, not a direct observation, and its reported upper/lower confidence bounds are not used as regression inputs. The surface is available only at annual resolution. The residual within-cluster exposure variation that actually identifies the historical coefficients (0.20–0.88 percentage points depending on country and product) is modest relative to the 10-point scale used for reporting (Section 5.6). Finally, Ethiopia's interview-date uncertainty (Section 5.9; Appendix C) is treated as an inherent limitation of that round's recoded date fields, not a design choice this thesis made.

6.6 What this thesis does not claim

This thesis does not claim that its findings establish a causal effect of WASH facilities on child growth, in either direction. It does not claim that its null or weak baseline associations are evidence that WASH investment is an ineffective use of development resources. It does not generalise findings from Ethiopia, Ghana, Kenya, and Nigeria to Sub-Saharan Africa as a region. It does not treat the historical extension's computational verification (Appendix B) as evidence of improved causal identification. And, as developed fully in Section 6.2, it does not treat the difference between the baseline's largely null results and the historical extension's one significant result as evidence that exposure timing, specifically, explains that difference, because the two samples differ in several other respects that this thesis's design cannot separate from timing. With these boundaries stated, the next section draws together what this thesis's evidence does, modestly, support.

7. Conclusion

This thesis examined the association between water and sanitation conditions and child anthropometric outcomes in four Sub-Saharan African countries — Ethiopia, Ghana, Kenya, and Nigeria — distinguishing throughout between a child’s own household facilities and the WASH conditions of the surrounding community, and between a single survey-date exposure measurement and a preliminary, geospatially matched measure of local WASH coverage during a child’s own early-life window.

The baseline evidence does not support a clear, robust association between improved household WASH and child height-for-age, weight-for-age, or weight-for-height once comparisons are restricted to households within the same DHS cluster: none of the six household cluster-fixed-effects coefficients across the three outcomes and two exposures reaches conventional significance under this thesis’s primary inference convention. Community-level WASH coverage shows a large, strongly significant association before adjustment, but this attenuates to statistical insignificance once socioeconomic status and region fixed effects are added — a pattern consistent with confounding by local development, though this design cannot establish confounding as the only explanation, and this thesis does not claim otherwise. These findings sit consistently alongside, rather than in contradiction to, the strongest available causal evidence on WASH interventions (Cumming et al.), which similarly finds no detectable effect of basic, household-level WASH interventions on linear growth, while being explicit that this does not challenge WASH’s biological plausibility and that a facility-type classification (this thesis’s own exposure) is not the same as a comprehensive intervention package (Section 2.3), so this parallel is suggestive rather than confirmatory.

The preliminary historical extension — linking three countries’ earlier DHS rounds to modelled annual local WASH coverage for a defined early-life window — produces one result that stands out from the rest of this thesis’s findings: a positive, statistically significant pooled association between local water coverage and height-for-age ($\beta = 1.03$ per 10 percentage points, 95% CI [0.51, 1.54], $p < .001$), independently reconstructed and confirmed computationally during this project. This result is reported with the full set of qualifications developed in Section 6: it is sensitive to the pooled weighting convention, rests on an undocumented measurement-subsample sampling weight, reflects area-averaged rather than population-weighted exposure around displaced cluster coordinates, and cannot be cleanly attributed to exposure timing alone because the baseline and historical samples differ in several respects simultaneously. It is a promising, carefully qualified preliminary finding, not a settled result, and this thesis does not use it to overturn or reinterpret the baseline’s more cautious conclusions.

Taken together, this thesis’s contribution is a transparent, side-by-side comparison of household and community WASH exposure within one consistent four-country framework, together with a bounded, honestly limited first attempt to introduce genuine temporal exposure variation into that comparison. Neither the baseline nor the historical results, individually or together, establish a causal effect of WASH facilities on child nutritional outcomes in these four countries, and neither should be read as evidence for or against the value of WASH investment as a development policy — a question this associational, four-country design was never positioned to answer. What this thesis does offer is a careful accounting of where the available evidence is strong, where it is weak, and where — particularly in the historical extension’s unresolved measurement documentation and the baseline’s inherent exposure-timing limitation — further data or a different design would be needed before a stronger claim could responsibly be made. Future work extending the historical approach to all four countries, resolving the measurement-

subsample weighting question through direct engagement with DHS methodological documentation, and pairing geospatial WASH matching with a genuinely panel or repeated-cross-section design, would be a natural next step toward narrowing the gap this thesis has tried to describe honestly rather than close.

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Appendix

Appendix A — Variable definitions, sample flow, and authoritative sources

A.1 Variable definitions

Variable	Definition	Source file (authoritative)
HAZ, WAZ, WHZ	Height-for-age, weight-for-age, weight-for-height z-scores, WHO Child Growth Standards reference (WHO Multicentre Growth Reference Study Group et al.)	DHS recode, flagged valid per DHS convention
Household water (binary)	JMP-improved household water source	docs/empirical_strategy.md §2
Household sanitation (binary)	JMP-improved household sanitation facility	docs/empirical_strategy.md §2
Community water/sanitation (water_rate_loo, sanitation_rate_loo_core)	Leave-one-out share of other households in the same cluster with the improved facility	docs/empirical_strategy.md §2
IDHSPSU	Sample-qualified DHS cluster code (unique across survey rounds)	IPUMS-DHS / DHS recode
PERWEIGHT	DHS/IPUMS standard child-level sampling weight	IPUMS-DHS documentation
W_IMP, S_IMP	IHME modelled annual local coverage of JMP-improved water/sanitation, 0–100, ~5×5 km, 2000–2017	docs/provenance/historical_wash_ihme_product_definitions_v1_2026-10-07.md
age_m (child's own age in months)	Reported completed months where delivered (Ethiopia, Nigeria baseline; Ethiopia, Nigeria historical); documented calendar-month difference otherwise (Ghana, Kenya)	docs/empirical_strategy.md; scripts/historical_wash/16_estimate_preliminary.py

Variable	Definition	Source file (authoritative)
b_year (birth year)	Child's birth year, used for survey-qualified country-by-birth-year fixed effects in the historical models	Derived in the analysis-sample construction stage; confirmed independent of the Ethiopia interview-date candidate (Appendix C)
cluster_key	Composite key encoding survey + cluster, sample-qualified by construction	data/processed/analysis_samples/run_20261006T184744Z/analysis_candidates_RESTRICTED.parquet (RESTRICTED; schema only cited here)

A.2 Baseline sample flow

Pooled harmonised child file (70,231 children, 4,481 clusters, 114 regions) → outcome-valid, measurement-universe-restricted samples (36,985 HAZ / 37,260 WAZ / 37,088 WHZ) → common samples requiring both household and community exposures classifiable (36,040 / 36,308 / 36,145, 4,426 clusters). Full flow: `outputs/household_community_wash/01_development_sample_flow.csv`.

A.3 Historical sample flow (established-date, post-12-month window, water, HAZ, as a worked example)

Raw analysis candidates, established countries (60,772) → `candidate == GREGORIAN_ESTABLISHED` (60,772; no further loss for the three established countries) → complete postnatal window + complete spatial coverage (42,052) → HAZ valid and within measured universe (23,135) → complete-case on covariates (23,135; no further loss) → singleton-cluster exclusion (−117 clusters, −117 observations) → **final: 23,018 children, 3,114 clusters**. Reproduced independently in `inscripts/historical_wash/20_validate_pooled_estimate.py`'s execution log (`data/processed/estimation/pooled_independent_validation_20261007T090003Z/execution_log.txt`, RESTRICTED).

A.4 DHS/IPUMS official documentation sources attempted for the measurement-subsample weight (Section 3.8, Section 6.3)

Source tried	Result
DHS-8 Biomarker Manual (dhsprogram.com PDF)	HTTP 403, tried twice across two sessions
Guide to DHS Statistics (dhsprogram.com)	HTTP 403
World Bank microdata catalogue, child anthropometry variable page	Accessible; no weighting guidance found
Nigeria 2018 NDHS final report	Accessible; describes sampling weights by stage/cluster and a domestic-violence special weight, but no anthropometry/biomarker-specific weight
IPUMS-DHS PERWEIGHT and KIDWT documentation pages	Accessible; confirms PERWEIGHT as the documented child-level tabulation weight; KIDWT is a population factor, not used

No source documents a measurement-subsample-specific weight. This is recorded as an unresolved documentation gap (issue register item #1), not inferred or substituted with an invented correction.

A.5 Full authoritative-file index

See docs/provenance/final_empirical_package_v3_2026-10-07/00_results_index.md and 02_core_reporting_table.md for the complete, current index of every baseline and historical output file this manuscript cites, with status labels (validated for main reporting / validated sensitivity / exploratory / invalid-superseded / not estimable).

Appendix B — Model specifications, inference conventions, and independent verification

B.1 Inference-convention comparison (baseline household cluster-FE model, HAZ)

Convention	Water β	SE	p	Sanitation β	SE	p
PRIMARY (same-group CRV1; this thesis’s main convention)	−0.085	0.046	.062	0.026	0.043	.548
FULL_DUMMY_DF sensitivity	−0.085	0.049	.080	0.026	0.046	.574

The two conventions differ only in whether the small-sample CRV1 correction additionally penalises degrees of freedom for the absorbed cluster dummies (FULL_DUMMY_DF does; PRIMARY does not, following the project’s cross-validated adjudication against `linearmodels.PanelOLS(auto_df=True)`; see `docs/provenance/cluster_fe_inference_adjudication.md`). Point estimates are identical by construction; only standard errors and p-values differ. Both are reported wherever this coefficient is cited in the main text, with the convention always named.

B.2 Independent verification of the pooled historical model (Section 4.6, Section 5.6)

This thesis’s own pipeline (`scripts/historical_wash/16_estimate_preliminary.py`) estimates all 291 historical models using a hand-written weighted within-cluster demeaning estimator. A prior independent check (`scripts/historical_wash/17_validate_estimates.py`) re-estimated four representative *country-specific* cells (Ghana water HAZ weighted and unweighted; Ghana sanitation WAZ; Nigeria sanitation HAZ) with an explicit-dummy-variable `statsmodels` implementation, matching to 3×10^{-14} .

This did not cover the pooled, multi-country model, which is the headline result of the historical extension. `scripts/historical_wash/20_validate_pooled_estimate.py`, written for this manuscript-preparation phase of the project, independently:

1. Rebuilds the estimation sample from the analysis-candidates stage directly (not from the stored, already-estimated model dataset), applying every filtering step explicitly and logging row counts (reproduced in Appendix A.3).
2. Estimates with `linearmodels.PanelOLS` — a different software library from the project’s own hand-written estimator — using entity (cluster) fixed effects and clustered CRV1 standard errors.
3. Reports both the PRIMARY (`auto_df=True`) and FULL_DUMMY_DF (`auto_df=False`) conventions.
4. Estimates the unweighted counterpart on exactly the same final observations.

First attempt and its correction, reported for transparency. An initial run of this independent script used a single pooled reference birth-year cell (rather than a separate reference year per country) and omitted the child’s own age-in-months as a linear control. This produced β per 10 points = 1.385 against the stored 1.026 — a roughly 35% discrepancy — despite the sam-

ple size (N=23,018), cluster count (G=3,114), and exposure mean/SD matching the stored values exactly, which is what indicated a specification difference rather than a sample-construction error. After matching the documented design exactly (country-local-reference birth-year dummies; child age-in-months included), the independent estimate converged to:

Quantity	Independent (corrected)	Stored	Absolute difference
β per 10 points	1.0255531	1.025553	6.7×10^{-8}
SE per 10 points (as originally run)	0.260694	0.260730	3.6×10^{-5}
N	23,018	23,018	0
G (clusters)	3,114	3,114	0
K (regressors) / rank	24 / 24	24 / 24	0
Residual exposure SD (points)	0.428348	0.428348	~0
Unweighted counterpart β per 10	0.4147	0.41 (documented)	consistent

Full execution log and comparison JSON: data/processed/estimation/pooled_independent_validation_20261007T090003Z/ (RESTRICTED; aggregate figures only, reproduced above).

The SE difference above is fully explained, not residual numerical noise. 16_estimate_preliminary.py's declared small-sample correction is $(G/(G-1)) \times ((N-1)/(N-K))$ applied to the raw cluster-robust sandwich, with CIs/p-values from a Student's t distribution on $G-1$ degrees of freedom (read directly from its source, lines 209–212). 20_validate_pooled_estimate.py's and 21_validate_all_primary_cells.py's PanelOLS.fit(cov_type="clustered", cluster_entity=True, auto_df=True) call applies only part of the equivalent correction: auto_df=True correctly sets the same-group, no-double-penalty degrees-of-freedom convention (confirmed by reading the installed linearmodels 7.0 source, panel/model.py's _determine_df_adjustment), and debiased=True (the library's own default) applies a $N/(N-K)$ scale — but the *separate*, opt-in group_debias=True flag, which applies linearmodels' own $(G/(G-1)) \times ((N-1)/N)$ cluster-count adjustment (shared/covariance.py, group_debias_coefficient), was not passed. Multiplying the two pieces linearmodels *did* apply, $[N/(N-K)] \times [(G/(G-1)) \times (N-1)/N]$, is algebraically identical to 16's declared $(G/(G-1)) \times (N-1)/(N-K)$ — so the missing flag, not a different formula and not floating-point noise, is the entire explanation. scripts/historical_wash/24_reconcile_inference_corrections.py (new) re-fits all 24 primary cells with group_debias=True added: **every one of the 24 cells' SE now matches the stored value to the displayed precision (max remaining difference 0.0 at 8 decimal places)**, and a direct algebraic check (uncorrected sandwich SE $\times$ the manually computed combined scalar vs. the library's own corrected SE) confirms a zero residual for all 24 cells. Full detail and the per-cell table: docs/provenance/independent_audit_v1_2026-10-07/09_update_2026-10-07_pass4.md. This does not change the stored, authoritative SE in 16_estimate_preliminary.py's own output anywhere — that estimator's formula was correct throughout; what was corrected is the *validation* scripts' covariance configuration, not the pipeline being validated.

What this verification establishes, and what it does not. It establishes that the reported pooled coefficient and its standard error are computationally correct — the numbers the documented specification, correctly implemented in a second, independent library with an equivalent covariance configuration, actually produce. It does not establish that the specification identifies

a causal effect, and it does not place this one cell in a higher *reporting* tier than its sibling historical models, all of which share the same unresolved documentation issues (Section 6; docs/provenance/historical_wash_issue_register_v1_2026-10-07.md).

B.3 Design checks applied to all 24 primary historical models

Record-key uniqueness, cluster-to-survey mapping (one survey per cluster), country weight totals equal to N/K within each model's own estimation sample, full design-matrix rank, and absence of singleton clusters in the final estimation data were checked programmatically for all 24 primary models (scripts/historical_wash/19_check_model_design.py; all checks pass, data/processed/estimation/run_20261006T185646Z/design_checks_20261006T190020Z/design_checks_summary.json, RESTRICTED).

Appendix C — Historical exposure-assignment diagnostics and the Ethiopia provisional analysis

Clearly labelled exploratory material, consistent with Section 5.9. Nothing in this appendix is part of the established-date pooled headline result and none of it should be cited as such.

C.1 Ethiopia’s three interview-date candidates

Ethiopian survey rounds record interview dates in the Ethiopian calendar, which must be converted to the Gregorian calendar to place a child’s postnatal exposure window relative to the (Gregorian-dated) IHME annual rasters. Three candidate conversions were constructed:

- **C1** (“triple,” 30-day Ethiopian-calendar arithmetic applied directly to the delivered day and month fields). **Invalid and superseded:** the delivered Ethiopian day and month fields are not themselves original 30-day Ethiopian dates (day value 31 occurs in months the Ethiopian calendar defines as having 30 days), so direct 30-day arithmetic conversion is invalid for these fields — a conclusion reached independently of, and prior to, any WASH-outcome analysis. 275 of 10,641 Ethiopia children have an outright invalid C1 day value under this method; the remainder share the same invalid conversion logic. C1 is excluded from all reporting tables in this thesis and its historical agreement with C2/C3 is not treated as evidence of either candidate’s validity. - **C2** (“century-day route,” origin 1 Meskerem 1900 = 12 September 1907). Provisional; not independently validated. - **C3** (“century-day route,” anchored 1 Meskerem 2008 = 12 September 2015, Gregorian cumulative month lengths). Provisional; not independently validated.

C.2 How much C2 and C3 actually agree (measured directly, not assumed)

Window	C2 eligible	C3 eligible	Disagree
Birth-year	9,862	9,862	0
Prenatal (9 months)	9,862	9,862	0
Post-12-month	7,930	7,922	8
Post-24-month	5,974	5,969	5

Out of 10,641 Ethiopia children: C1-vs-C2/C3 differ by exactly the 275 invalid-day exclusions (and C2/C3 never differ from each other on birth year itself — identical across all three candidates for every child, since birth year is derived from the birth-date fields, not the interview date). This supports two precise, narrow claims: (1) C2 and C3 agree almost perfectly on window eligibility, so the remaining uncertainty between them changes very few children’s classification; (2) this is *not* evidence that either candidate’s absolute date is correct, nor that birth-year exposure can be established entirely without a valid interview-date conversion, since even the eligibility check itself (not only the postnatal windows) requires a working Gregorian interview date to confirm the window has been completed by the time of interview.

C.3 Ethiopia provisional coefficients (water HAZ, sanitation HAZ, post-12-month window)

Candidate	Water β per 10	Sanitation β per 10
C2	-0.39	-0.45
C3	-0.39	-0.45

Both confidence intervals include zero. These are reported here, in this appendix only, as an exploratory sensitivity with no bearing on the established-date pooled results in Section 5.6.

C.4 Spatial-extraction and sensitivity diagnostics (established-date models, summarised)

Diagnostic	Result
Buffer coverage-completeness tolerance	Within 10^{-3} of full buffer area
Spatial-method sensitivity (containing-pixel vs. exact buffer average)	Pooled water-HAZ: 0.68 vs. main 1.03
Rural-10km buffer sensitivity (urban unchanged)	Pooled water-HAZ: 1.01 vs. main 1.03
Stricter postnatal completion rule (+1 month)	Pooled water-HAZ: 0.95 vs. main 1.03
Calendar-month age construction throughout	Pooled water-HAZ: 1.03 (materially unchanged)
DHS/IHME overlap exclusion (Ghana removed)	Pooled water-HAZ: 0.74, Kenya+Nigeria only

No sensitivity cell was selected or suppressed based on its significance; all are reported together as a pre-specified sensitivity family (docs/provenance/historical_wash_preliminary_estimation_methods_2026-10-06.md §7).

I hereby affirm that I have written the above thesis independently and have used no sources or aids other than those indicated; that the submitted work has not previously been submitted for examination at any other university; and that it has not been published, either in whole or in part. I have clearly indicated all direct quotations and passages paraphrased from other works as such.

Violations of this declaration may be considered an attempt at deception.

Place, Date

Signature